Research Proposal

Course: IS 204: Research Design in Information Sciences

Title: The role of AI in recruiting in Tech

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Artificial intelligence is changing the way many companies recruit job candidates, especially in the technology industry. More employers are using AI to screen resumes and evaluate applicants, but this raises questions about how fair and reliable these systems really are. This study looks at how AI is affecting the hiring process and whether it actually improves efficiency and fairness or creates new problems. The goal is to understand if using AI in recruitment is beneficial and should continue, or if human screening may lead to less biased decisions. To explore this topic, the study will use both quantitative and qualitative methods. Hiring data will be analyzed to identify patterns in who gets selected or rejected, particularly across different demographic groups. In addition, interviews with recruiters and job seekers will help capture personal experiences and opinions about AI in hiring. Overall, this research aims to better understand AI's role in recruitment and contribute to more fair and transparent hiring practices.

In recent years, artificial intelligence has become more common in the way organizations operate, especially within recruitment. Many companies now rely on AI driven tools to automate time consuming tasks such as resume screening, candidate ranking, and interview scheduling. These tools are often used to speed up the hiring process and manage a large applicant pool more efficiently. While AI has the potential to save time and reduce costs, it also raises many concerns about its fairness and transparency in hiring decisions. Some organizations choose not to use AI at all for recruiting and rely on human judgment instead. This issue is especially important in the technology industry, where there is a high demand for skilled workers and hiring decisions can strongly affect both the companies and applicants. Despite the increasing use of AI in recruitment, there is not as much understanding of how these technologies impact hiring outcomes specifically within the technology industry. Many companies adopt AI tools to improve efficiency, but concerns remain about algorithmic bias and unequal opportunities for certain candidates. This lack of clarity highlights the need to examine whether AI-driven recruitment systems truly create fair outcomes or unintentionally disadvantage certain groups.

The purpose of this study is to examine how artificial intelligence influences the recruitment process for technology roles, with a focus on efficiency, accuracy, and potential bias in candidate selection. This research aims to understand whether AI improves hiring decisions or creates new challenges. Specifically, the study explores how the use of AI has changed the recruitment process for technology roles and how these changes affect both employers and applicants. It also looks at the extent to which AI impacts fairness and bias in hiring decisions within the technology industry. Another key focus of the study is comparing AI based resume screening to traditional human resume screening in terms of accuracy and effectiveness. By addressing these questions, the study hopes to provide insight into whether AI should continue to play a role in recruiting or if adjustments should be made to ensure more fair hiring practices.

For this study, the Technology Acceptance Model (TAM) is used as the primary theoretical framework. TAM explains how individuals decide whether to adopt new technologies based on two factors, perceived usefulness and perceived ease of use. This framework is particularly relevant to AI-based hiring tools, as recruiters are more likely to adopt these systems if they believe the tools save time, improve decision making, or increase efficiency. Conversely, if AI systems are perceived as difficult to use or unreliable, recruiters may resist their adoption. TAM also emphasizes that user perceptions shape how technology is implemented in practice, which is critical when examining AI's role in recruitment. By applying this model, the study can better understand how recruiters and organizations accept or reject AI tools and how these attitudes influence hiring outcomes. The model will also guide data collection and analysis by helping explain why trust in AI varies among recruiters and job candidates.

Rodney, Valášková, and Durana (2019) examine how artificial intelligence and automation enhance recruitment efficiency and support decision-making processes while also raising concerns related to ethics, privacy, and fairness. Although their findings highlight organizational benefits such as reduced hiring time and increased consistency, the study largely overlooks candidate experiences, algorithmic bias, and transparency. Additionally, the research adopts a broad, cross-industry perspective, offering limited insight into how AI-driven recruitment specifically operates within the technology sector. This gap underscores the need for industry-focused research that considers both organizational outcomes and applicant perspectives.

` Abdelhay (2023) analyzes the role of AI in transforming recruitment and human resource operations through automated screening, reduced administrative costs, and improved hiring quality. The study portrays AI as a tool that enhances productivity, minimizes bias, and improves decision-making accuracy. However, the research relies heavily on secondary data and provides limited critical discussion of potential risks, including algorithmic bias, data privacy concerns, and candidate perceptions. Furthermore, it does not sufficiently address how these issues manifest within technology-sector hiring, where AI adoption is particularly prevalent.

Gallegos (2024) explores public attitudes toward the use of AI in hiring through a collection of reader responses, revealing both optimism and skepticism. Some respondents view AI as a fairer and more consistent alternative to human decision-making, while others argue that it may dehumanize the hiring process, reinforce discrimination, or overlook interpersonal qualities that algorithms cannot assess. Several contributors emphasize the importance of using AI as a supportive tool rather than a primary decision-maker, highlighting the need for transparency and explainability to build trust. This article reflects an ongoing debate about balancing efficiency, fairness, and human judgment in recruitment.

These studies show that even though AI recruitment shows both efficiency and consistency there are definitely concerns regarding fairness, transparency, and trust. Existing literature focuses on organizational benefits or public opinion, with not a lot of attention to the technology part specifically or to comparative perspectives between recruiters and job seekers. This study addresses these gaps by looking into AI driven hiring within the technology industry and putting in both quantitative hiring outcomes and qualitative experiences to better understand how automation influences fairness, accuracy, and efficiency in recruitment.

The objectives of this study are to examine how artificial intelligence tools are used in the hiring process for technology related jobs, including resume screening, candidate matching, and interview scheduling. This study also aims to explore how the use of AI changes the role of human recruiters and whether it improves the speed and efficiency of the hiring process. Additionally, this research looks to analyze the impact of AI on fairness and bias in recruitment, with attention to whether AI systems help reduce discrimination or reinforce existing biases in technology sector hiring.

This study will follow basic ethical research guidelines throughout the entire research process. All the participants will be asked to give informed consent before taking part in the study. This will include a clear explanation of the study's purpose, what participation involves, and the option to withdraw at any time without any penalty. Participation will be completely voluntary. To protect privacy, no names or identifying information will be collected. Instead, participants will be assigned unique identifiers so their responses can be anonymous. All collected data will be secured and will only be accessible to the team. Since the study involves personal experiences related to hiring and recruitment, participants can choose to skip any questions that make them uncomfortable. The research team will take care to design survey and interview questions using neutral language to reduce bias. Overall, ethical considerations will be prioritized to protect participants and ensure that the findings are collected and reported responsibly.

This study uses a mixed-methods research design to examine the impact of artificial intelligence on recruitment in the technology industry. Quantitative data will be used to analyze hiring-related trends such as time-to-hire, selection outcomes, and perceived fairness, while qualitative data will capture the experiences and perceptions of recruiters and job seekers who have interacted with AI-based hiring tools. Participants will include recruiters, hiring managers, and job seekers from the technology industry and will be recruited through professional networking platforms, online forums, and university connections. Data will be collected through online surveys, semi-structured interviews, and publicly available datasets and industry reports. Surveys will include multiple-choice and Likert-scale questions, and interviews with approximately 10–15 participants will explore perceptions of efficiency, fairness, and bias in greater depth. Quantitative data will be analyzed using descriptive statistics, while qualitative data will be analyzed using thematic analysis, with a small pilot study conducted beforehand to refine the research instruments.

There are a few potential risks in this study. Some participants may feel uncomfortable sharing personal or sensitive information, the sample may not be fully diverse, and there is also a chance of unintentional bias in how the data is interpreted. To reduce these risks, participants will be informed that only the researchers will have access to the data and that they may skip any questions they do not feel comfortable answering. We will also try to recruit a larger and more diverse group of participants to better represent different backgrounds and experiences. To reduce researcher bias, survey and interview questions will be written using clear, neutral language. We will also be transparent throughout the research process to make sure conclusions are based on the data rather than personal opinions. Participant data will be labeled using identifiers instead of names and stored in secure locations. The final paper will only include summarized results that cannot be traced back to individual participants.

Through this study, we expect to find that while AI can make the hiring process faster, it also creates concerns about fairness and bias. The quantitative results will likely show that AI-based hiring tools reduce overall hiring time, but differences across demographic groups may still appear. For example, AI systems may favor applicants from well-known or highly ranked schools, even though many qualified candidates attend less prestigious schools due to financial or personal circumstances. Interview responses are expected to show mixed opinions about using AI in hiring. Some participants may see AI as efficient and helpful, while others may view it as biased or unfair. Overall, these findings can help improve understanding of how people interact with AI tools in hiring decisions. This study uses a sociotechnical perspective, meaning that technology and society influence each other.

This study has several limitations, including time constraints, limited access to hiring data, and sampling challenges. Because the research is conducted within a single semester, the scope is limited to a manageable number of participants and data sources. Access to detailed company hiring data may also be restricted due to privacy concerns, which could require greater reliance on survey data. To address these limitations, the study will use publicly available datasets, prior research, company transparency reports, and survey responses from recruiters and job seekers when available.

Overall, this study aims to better understand how artificial intelligence is changing the hiring process, particularly in terms of efficiency, fairness, and bias. By combining surveys, interviews, and publicly available data, the research highlights both the benefits and challenges of AI-driven recruitment. Despite existing limitations, the flexible study design allows meaningful insights to be drawn and contributes to discussions about using AI responsibly in hiring decisions.

References

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